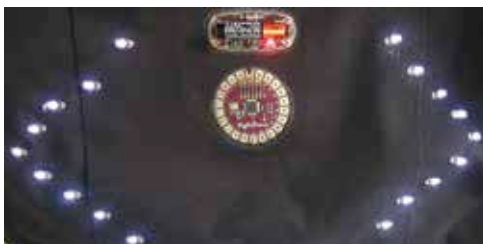


you may want to build your own. While it's possible using only switches and LED lights, there's an added advantage that it could be turned off automatically in case you forget to turn it off.

What's more, many Arduino™ board circuits could even be embedded into your clothing!



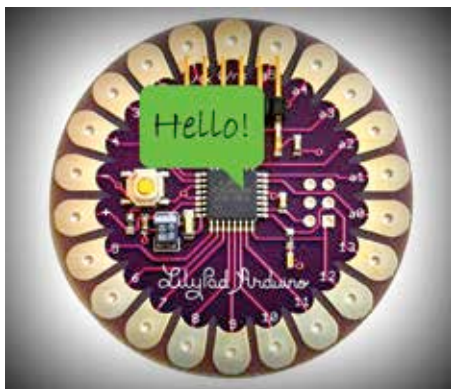
Bicycle turn indicator

IMAGE CREDIT:
CYLLIGHTSWORDPRESS.COM

Say hello to the Arduino™...

The Arduino™ is a combined package of a Printed Circuit Board (commonly known as 'PCB', and what we've been referring to all along as a 'microcontroller') and software program called Arduino™ that helps give the board instructions.

The PCB has all the circuitry that makes the microcontroller on your Arduino™ board ready to use. This includes a crystal oscillator that creates a clock signal, which decides the speed of operation of the microcontroller.



An Arduino™ board

The Arduino™ software compiles (converts to language of microcontroller) the program you flash (upload) to the microcontroller.

To use a microcontroller, we first need you to understand the architecture and functions built into it. Ask any engineering student to use a microcontroller and the first question in their minds will be: "From where do we flash it and burn the code?" As you'll learn later, programming for even small tasks requires the knowledge of registers, clocks, timers, interrupts, programming headers etc.